

SOFTWARE REQUIREMENTS SPECIFICATION

CampusRide

Smart College Bus Tracking & Management System

Software Requirements Specification (SRS) Document

Document Information

Item	Details
Project Name	CampusRide – Smart College Bus Tracking & Management System
Document Type	Software Requirements Specification (SRS)
Target Platform	Android Mobile Application and Web-based Admin Dashboard
Primary Users	Students, Drivers, and Transport Administrators
Technology Stack	Flutter, Firebase, Google Maps API, Android Studio

1. Introduction

1.1 Purpose

CampusRide is a smart college bus tracking and management system designed to provide real-time tracking of college buses. The system enables students to view the current location and estimated time of arrival (ETA) of their assigned bus. It also enables drivers to share their live GPS location and allows transport administrators to manage buses, drivers, students, routes, notifications, and reports efficiently.

1.2 Problem Statement

Students often wait at bus stops without knowing the exact location or arrival time of the college bus. Drivers may currently share live locations manually through messaging applications, which can be unreliable and may be forgotten. This results in unnecessary waiting, delays, confusion, and poor communication between students, drivers, and the transport administration.

1.3 Objectives

- Provide real-time tracking of college buses.
- Reduce student waiting time at bus stops.
- Provide estimated time of arrival (ETA) information.
- Improve communication between students, drivers, and transport administrators.
- Enable efficient management of college transport operations.
- Improve safety and transparency in college transportation.

2. Overall Description

2.1 Product Perspective

CampusRide will function as a centralized transport management system consisting of an Android mobile application for students and drivers and a web-based dashboard for administrators. The system will use Firebase services for authentication, database management, cloud communication, and notifications. Google Maps API will be used for map visualization and location tracking.

2.2 User Classes

User	Role	Main Responsibilities
Student	Bus passenger	Track assigned bus, view ETA, check routes/stops, mark attendance status, receive notifications
Driver	Bus operator	Start/end trips, share GPS location, view assigned students, manage trip status.
Admin	Transport administrator	Manage buses, drivers, students, assignments, monitoring, notifications, and reports.

2.3 Platform

- Android Mobile Application for Students and Drivers.
- Web Dashboard for Transport Administrators.
- Cloud-based backend infrastructure.

3. Functional Requirements

3.1 Student Module

- 1 The student shall be able to log in using a valid College ID and password.
- 2 The student shall be able to view the bus assigned to them.
- 3 The student shall be able to track the assigned bus on a live map.
- 4 The student shall be able to view the estimated time of arrival (ETA).
- 5 The student shall be able to view the bus route and designated stops.
- 6 The student shall be able to mark their status as 'Coming Today' or 'Not Coming Today'.
- 7 The student shall receive important transport notifications.
- 8 The student shall be able to access an emergency call feature.

3.2 Driver Module

- 1 The driver shall be able to securely log in to the application.
- 2 The driver shall be able to start a bus trip.
- 3 The system shall share the driver's GPS location during an active trip.
- 4 The driver shall be able to view the list of students assigned to the bus.
- 5 The driver shall be able to view students marked as 'Coming Today'.
- 6 The driver shall be able to end the trip after reaching the destination.

3.3 Admin Module

- 1 The administrator shall be able to securely log in to the dashboard.
- 2 The administrator shall be able to add, edit, and delete bus records.
- 3 The administrator shall be able to add and manage driver accounts.
- 4 The administrator shall be able to add and manage student records.
- 5 The administrator shall be able to assign students to buses.
- 6 The administrator shall be able to monitor active buses on a map.
- 7 The administrator shall be able to send notifications to users.
- 8 The administrator shall be able to generate transport-related reports.

4. Non-Functional Requirements

Requirement	Description
Security	The system shall provide secure authentication and protect user data from unauthorized access.
Performance	The system should provide fast response times for common operations such as login, bus tracking, and data retrieval.

Requirement	Description
Accuracy	The system should provide accurate GPS location information whenever a driver is on an active trip.
Usability	The application should have a simple, clear, and user-friendly interface.
Reliability	The system should remain available and function reliably during college transport operations.
Scalability	The system should support the addition of more buses, students, drivers, and routes in the future.
Privacy	Personal information and location data should be handled securely and only be accessible to authorized users.

5. Hardware Requirements

5.1 User Devices

- Android smartphone.
- GPS-enabled device.
- Active internet connection.

5.2 Server Infrastructure

- Cloud server infrastructure.
- Firebase cloud services for backend operations and database management.

6. Software Requirements

Software / Technology	Purpose
Flutter	Development of the Android mobile application.
Firebase Authentication	User authentication and secure login.
Firebase Firestore	Cloud database for storing application data.
Google Maps API	Map display, route visualization, and location tracking.
Firebase Cloud Messaging	Push notifications.
Android Studio	Application development and testing environment.

7. System Architecture

The CampusRide system follows a cloud-based client-server architecture. The Student App and Driver App communicate with Firebase Cloud Services. Firebase manages authentication, database operations, and notifications. Google Maps API is used to display real-time bus locations and routes. The Admin Dashboard provides centralized management and monitoring.

Student App



Driver App ■■■■■■■■ Firebase Cloud

■ (Authentication + Database)



GPS Location Admin Dashboard



Google Maps API

8. Use Cases

Student

- Login
- View Assigned Bus
- Track Bus
- View ETA
- View Route and Stops
- Mark Coming / Not Coming
- Receive Notifications
- Emergency Call

Driver

- Login
- Start Trip
- Share GPS Location
- View Student List
- View Students Coming Today
- End Trip

Admin

- Login
- Manage Students
- Manage Drivers
- Manage Buses
- Assign Students to Buses
- Monitor Live Buses
- Send Notifications
- Generate Reports

9. Data Requirements

The system is expected to maintain the following major data entities:

- Student: Student ID, name, contact information, assigned bus, route, and attendance status.
- Driver: Driver ID, name, contact information, login credentials, and assigned bus.
- Bus: Bus ID, registration number, capacity, assigned driver, route, and current status.
- Route: Route ID, route name, stops, and route information.
- Location: Bus ID, GPS coordinates, timestamp, and trip status.
- Notification: Message, recipient, date/time, and notification status.

10. Assumptions and Constraints

- Users must have a valid internet connection to access real-time tracking features.
- The driver's device must have GPS enabled during an active trip.
- Location accuracy may vary depending on GPS signal and network availability.
- Only authorized users can access role-specific features.
- The system depends on the availability of Firebase and Google Maps services.

11. Future Scope

- Parent tracking facility.
- AI-based ETA prediction.
- QR Code-based boarding attendance.
- Real-time seat availability.
- Traffic-based route optimization.
- SOS emergency feature.
- Multi-college and multi-campus support.
- Advanced analytics for transport management.

12. Expected Outcome

CampusRide is expected to provide a reliable, user-friendly, and efficient college transport management system. The system will minimize student waiting time, improve communication between students and transport staff, enhance transport safety and visibility, and simplify the management of college bus operations.

13. Suggested Team Module Distribution

Team Member	Module / Responsibility
Member 1	Student Module
Member 2	Driver Module
Member 3	Admin Dashboard
Member 4	Firebase, Google Maps Integration, Notifications, and Backend

14. Conclusion

The CampusRide system addresses the common problems associated with college bus transportation by combining real-time GPS tracking, cloud-based data management, notifications, and administrative controls into a single platform. By connecting students, drivers, and transport administrators, the system aims to make college transportation more transparent, efficient, safe, and convenient.